

Genomics Club



Marwadi
University
Marwadi Chandarana Group



Coordinator:

Dr Aditya Saxena

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Cabin: MA557/Ext:1568

Department of Bioinformatics

Faculty of Engineering and Technology

Membership:

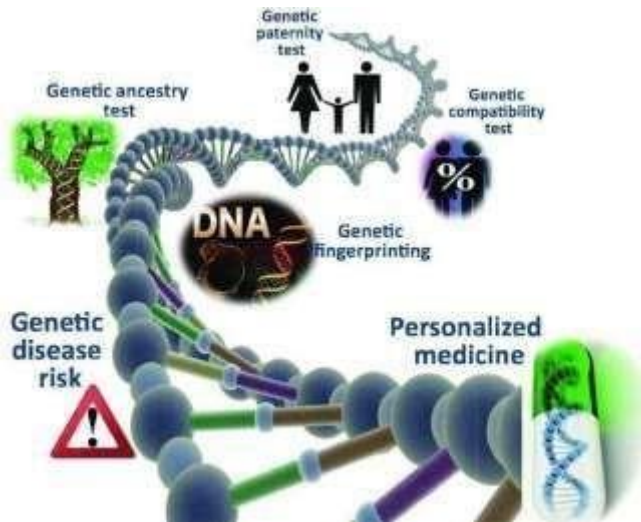
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Genomics club**

Vision: To enable students to explore the latest genomic technologies shaping biomedical research and applications.

Mission:

- Organize seminars, workshops, quizzes, games, and other recreational activities focused on Genomics technologies and applications.
- Foster curiosity and encourage the exploration of new knowledge about the latest Genomics technologies.
- Through organized events, share relevant information, knowledge, and skills necessary for achieving success in the professional world.

What is Genomics?



- The genome is the entire set of genetic instructions found in a cell. In humans, the genome consists of 23 pairs of chromosomes found in the nucleus and a small chromosome found in the cells' mitochondria. Each set of 23 chromosomes contains approximately 3.1 billion bases of DNA sequence.
- Genomics is an interdisciplinary branch of molecular biology concerned with the structure, function, evolution, and mapping of genomes. It's a rapidly evolving science that can potentially reduce global health inequalities by providing developing countries with efficient, cost-effective, and robust means of preventing, diagnosing, and treating major diseases that burden their populations. Genomic-based healthcare involves using genomic information about an individual for clinical decisions (i.e., for diagnostic or therapeutic decision-making) and considering the implications of that use.
- Genomics research offers the long-term possibility of providing new approaches for the prevention and management of many intractable diseases. It's important to prepare society for the complexities of this new field, to ensure that its benefits are distributed fairly among the countries of the world and that the more conventional approaches to medical research and practice are not neglected. In contrast, the medical potential of genomics is being explored.

Benefit to Students:

1. Ability to understand the significance and potential in learning and utilizing the wide variety of Genomics technologies and applications.
2. Gain an insight into the fast-evolving and thriving field of Genomics technologies and applications with the advantage of opening up new career choices and prospects.
3. Feel empowered and proficient by gathering information, knowledge and skills required to achieve professional success in future.
4. Opportunity to broaden their horizon and enrich their knowledge about new and improved Genomics techniques.
5. Used in clinical domain to make informed choices about the health issues in their family.
6. Practical exposure, industry insights, and networking opportunities, enhancing their knowledge and career prospects in biotechnology/bioinformatics.

Activities:

1. **Genomics:** Learn about the central dogma of molecular biology and how the DNA code found in all living things impacts our world.
2. **Applications of Genomics:** Explore the various applications of Genomics-based research and technologies.
3. **Cutting-Edge Techniques:** Find out more about the cutting-edge tools and techniques used in Genomics.
4. **Human Genome Project:** Understand the impact of the Human Genome Project on healthcare and various scientific disciplines.
5. **Explore Your Favorite Species:** Research your favorite animal or plant and create a PowerPoint presentation to share your findings with the club.
6. **DNA and Adaptations:** Discover how DNA sequences code for proteins, giving animals or plants different adaptations for survival.
7. **Disease Research Presentation:** Choose a disease to research and present your findings through a PowerPoint presentation.
8. **Microbes and Our World:** Explore the marvelous world of microbes and their impact on our lives.
9. **Microbes and DNA:** Learn how DNA can provide clues to how microbes work and whether they are 'good' or 'bad'.
10. **Career Exploration:** Explore the diversity of career pathways related to Genomics technologies.
11. **Ethical Debates:** Engage in group discussions and debates on current and potential ethical issues in genetics and genomics.
12. **Industrial Visit:** Visit to biotechnology/bioinformatics research centres/industries to provide practical exposure and industry insights to enhance the knowledge in the field of bioinformatics and biotechnology.

Membership:

Membership fee INR 100

Students/Faculty can submit fees at MA557

Club-Coordinator:

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Student Coordinators:

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THANK YOU